Part 1 (Saif): Network Design & RIP Configuration

Oct 11, 2026 · @saif

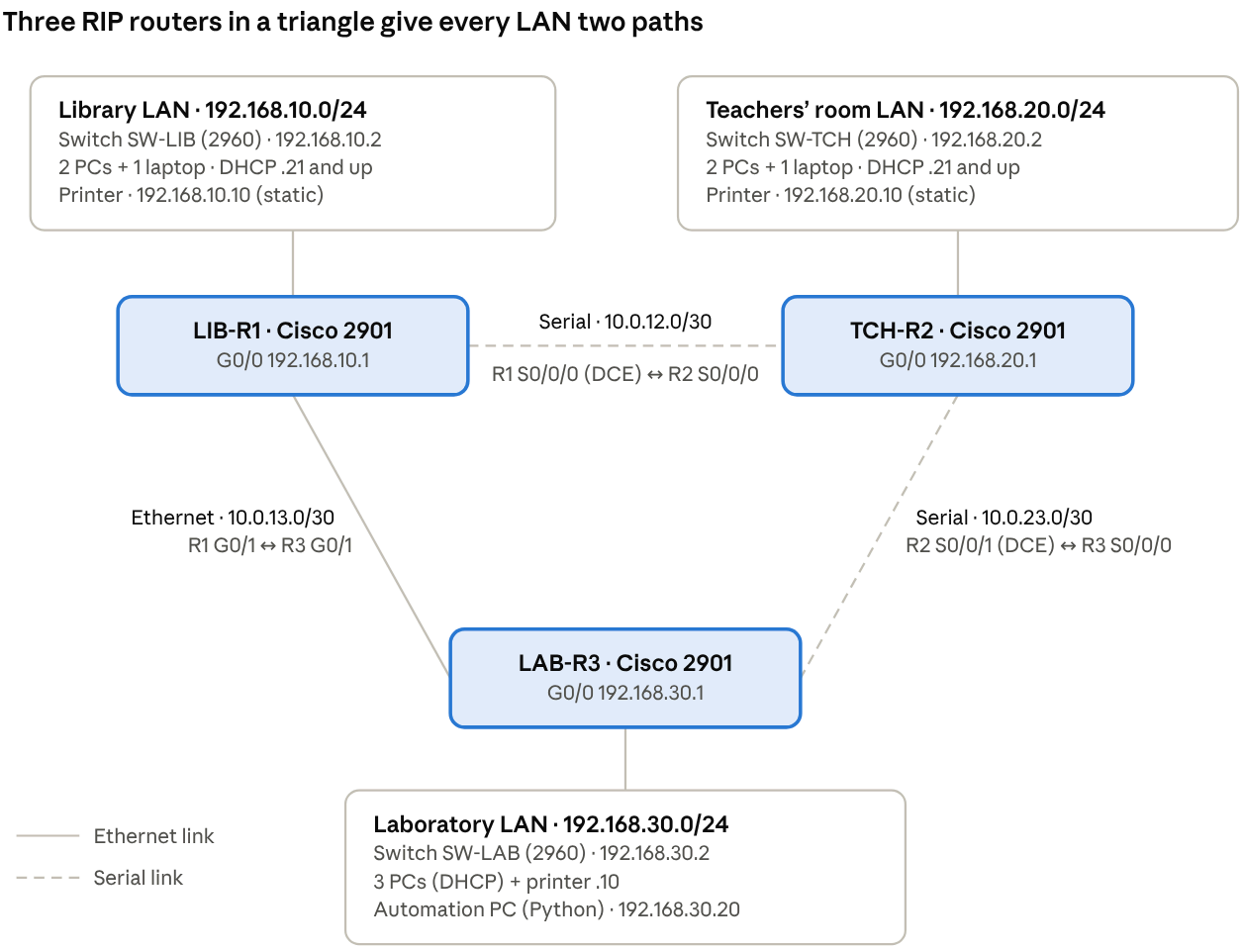
Scope

Saif builds the network itself: topology, IP plan, cabling and the full Cisco configuration, with RIPv2 running and verified on all three routers. Abdulrahman's Python scripts and Mouhamed's failure tests both run on top of this build, so this part comes first.

This part delivers, for the report and the repo:

- Topology diagram (Library, Teachers' room, Laboratory)
- IP addressing plan for every interface and device
- Configs for 3 routers and 3 switches, also as `.txt` files for the repo
- Build steps for Packet Tracer or the physical lab
- Verification checklist with the expected routing tables (the healthy-state snapshot)
- SSH access so the Python scripts can log in

Topology



School network · 3 routers, 3 switches, 3 router-to-router links

The three routers form a triangle, so every room can reach every other room two ways: directly, or through the third router. That gives the two redundant paths the brief asks for, and it is what makes the link-failure tests possible.

- R1-R3 is an Ethernet link and the other two are serial links, so the build uses both link types.
- Each router's G0/0 goes to its room's switch; the switch connects the PCs, laptops and printer.
- All three routers run RIPv2 with auto-summary off and advertise their LAN plus their WAN links.

IP addressing plan

Each room gets its own /24 LAN and each router-to-router link gets a /30 (exactly 2 usable addresses). Inside every LAN the same rule applies: .1 router, .2 switch, .10 printer, .20 automation PC (Lab only), DHCP hands out .21 and up.

Router interfaces

Router	Interface	IP address	Mask	Connects to
LIB-R1 (Library)	G0/0	192.168.10.1	255.255.255.0	SW-LIB Gi0/1
LIB-R1 (Library)	G0/1	10.0.13.1	255.255.255.252	LAB-R3 G0/1 (Ethernet)
LIB-R1 (Library)	S0/0/0	10.0.12.1	255.255.255.252	TCH-R2 S0/0/0 (Serial, DCE end)
TCH-R2 (Teachers)	G0/0	192.168.20.1	255.255.255.0	SW-TCH Gi0/1
TCH-R2 (Teachers)	S0/0/0	10.0.12.2	255.255.255.252	LIB-R1 S0/0/0 (Serial, DTE end)
TCH-R2 (Teachers)	S0/0/1	10.0.23.1	255.255.255.252	LAB-R3 S0/0/0 (Serial, DCE end)
LAB-R3 (Lab)	G0/0	192.168.30.1	255.255.255.0	SW-LAB Gi0/1
LAB-R3 (Lab)	G0/1	10.0.13.2	255.255.255.252	LIB-R1 G0/1 (Ethernet)
LAB-R3 (Lab)	S0/0/0	10.0.23.2	255.255.255.252	TCH-R2 S0/0/1 (Serial, DTE end)

Switches and end devices (mask 255.255.255.0 for all)

Room	Device	IP address	Gateway	Switch port	Set by
Library	SW-LIB (VLAN 1)	192.168.10.2	192.168.10.1	n/a	Static
Library	PC-LIB-1, PC-LIB-2, Laptop-LIB-1	192.168.10.21 and up	192.168.10.1	Fa0/1 to Fa0/3	DHCP
Library	Printer-LIB	192.168.10.10	192.168.10.1	Fa0/4	Static
Teachers' room	SW-TCH (VLAN 1)	192.168.20.2	192.168.20.1	n/a	Static

Room	Device	IP address	Gateway	Switch port	Set by
Teachers' room	PC-TCH-1, PC-TCH-2, Laptop-TCH-1	192.168.20.21 and up	192.168.20.1	Fa0/1 to Fa0/3	DHCP
Teachers' room	Printer-TCH	192.168.20.10	192.168.20.1	Fa0/4	Static
Laboratory	SW-LAB (VLAN 1)	192.168.30.2	192.168.30.1	n/a	Static
Laboratory	PC-LAB-1, PC-LAB-2, PC-LAB-3	192.168.30.21 and up	192.168.30.1	Fa0/1 to Fa0/3	DHCP
Laboratory	Printer-LAB	192.168.30.10	192.168.30.1	Fa0/4	Static
Laboratory	AUTO-PC (runs the Python scripts)	192.168.30.20	192.168.30.1	Fa0/5	Static

Hardware and cabling

A Cisco 2901 has only two Ethernet ports (G0/0 and G0/1), so each router needs an **HWIC-2T** serial module to get the extra ports. With the module in slot 0 the router gains Serial0/0/0 and Serial0/0/1.

Equipment	Qty	Notes
Cisco 2901 router	3	One per room; add an HWIC-2T to each (power off, insert, power on)
Cisco 2960 switch	3	One per room; 24 FastEthernet ports + Gi0/1 uplink
PCs / laptops	9	3 per room
Printers	3	1 per room
Automation PC	1	In the Lab; runs the Python scripts

Link	Ends	Cable type
Router to its switch (x3)	Router G0/0 to switch Gi0/1	Copper straight-through
PCs, laptops, printers to switch	Device NIC to switch Fa0/1, Fa0/2, …	Copper straight-through
Library to Lab	LIB-R1 G0/1 to LAB-R3 G0/1	Copper cross-over

Link	Ends	Cable type
Library to Teachers	LIB-R1 S0/0/0 to TCH-R2 S0/0/0	Serial DCE (DCE at LIB-R1)
Teachers to Lab	TCH-R2 S0/0/1 to LAB-R3 S0/0/0	Serial DCE (DCE at TCH-R2)

The DCE end of a serial cable supplies the clock, which is why only LIB-R1 S0/0/0 and TCH-R2 S0/0/1 carry `clock rate 64000`. In Packet Tracer, the router you click **first** with the Serial DCE cable becomes the DCE end. In the physical lab you also need a console (rollover) cable to configure each device the first time.

Router configurations

Each config is pasted whole into the router's CLI and does four jobs: interfaces, DHCP for the room, RIPv2, and SSH for the Python scripts. The same text is in the `.txt` files (LIB-R1, TCH-R2, LAB-R3) for the repo.

What the RIP block does, line by line:

- `router rip` starts the RIP process.
- `version 2` switches to RIPv2: subnet masks travel with each route and updates go to multicast 224.0.0.9.
- `no auto-summary` stops the router collapsing 10.0.12.0/30, 10.0.13.0/30 and 10.0.23.0/30 into one 10.0.0.0/8 route.
- `network 10.0.0.0` turns RIP on for every interface in 10.x.x.x (the two WAN links) and advertises them. RIP network statements are classful, so one line covers all three /30s.
- `network 192.168.10.0` (20.0 / 30.0 on the others) advertises the room's LAN.
- `passive-interface GigabitEthernet0/0` stops RIP updates going out to the PCs, while still advertising the LAN to the other routers.

LIB-R1 (Library)

```
enable
configure terminal
hostname LIB-R1
no ip domain-lookup
service password-encryption
enable secret class123
```

```
banner motd # LIB-R1 - Library router. Authorized access only. #
!
interface GigabitEthernet0/0
 description LAN - Library switch SW-LIB Gi0/1
 ip address 192.168.10.1 255.255.255.0
 no shutdown
 exit
interface GigabitEthernet0/1
 description WAN Ethernet - to LAB-R3 G0/1
 ip address 10.0.13.1 255.255.255.252
 no shutdown
 exit
interface Serial0/0/0
 description WAN Serial - to TCH-R2 S0/0/0 (this end is DCE)
 ip address 10.0.12.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit
!
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp pool LIB-LAN
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 exit
!
router rip
 version 2
 no auto-summary
 network 10.0.0.0
 network 192.168.10.0
 passive-interface GigabitEthernet0/0
 exit
!
ip domain-name school.local
username admin privilege 15 secret Admin@123
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
line console 0
 password cisco
 login
 logging synchronous
 exit
line vty 0 4
```

```
 login local
 transport input ssh
 exit
end
write memory
```

TCH-R2 (Teachers' room)

```
enable
configure terminal
hostname TCH-R2
no ip domain-lookup
service password-encryption
enable secret class123
banner motd # TCH-R2 - Teachers room router. Authorized access only. #
!
interface GigabitEthernet0/0
 description LAN - Teachers switch SW-TCH Gi0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown
 exit
interface Serial0/0/0
 description WAN Serial - to LIB-R1 S0/0/0 (this end is DTE)
 ip address 10.0.12.2 255.255.255.252
 no shutdown
 exit
interface Serial0/0/1
 description WAN Serial - to LAB-R3 S0/0/0 (this end is DCE)
 ip address 10.0.23.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit
!
ip dhcp excluded-address 192.168.20.1 192.168.20.20
ip dhcp pool TCH-LAN
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 exit
!
router rip
 version 2
 no auto-summary
 network 10.0.0.0
```

```
 network 192.168.20.0
 passive-interface GigabitEthernet0/0
 exit
!
ip domain-name school.local
username admin privilege 15 secret Admin@123
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
line console 0
 password cisco
 login
 logging synchronous
 exit
line vty 0 4
 login local
 transport input ssh
 exit
end
write memory
```

LAB-R3 (Laboratory)

```
enable
configure terminal
hostname LAB-R3
no ip domain-lookup
service password-encryption
enable secret class123
banner motd # LAB-R3 - Laboratory router. Authorized access only. #
!
interface GigabitEthernet0/0
 description LAN - Lab switch SW-LAB Gi0/1
 ip address 192.168.30.1 255.255.255.0
 no shutdown
 exit
interface GigabitEthernet0/1
 description WAN Ethernet - to LIB-R1 G0/1
 ip address 10.0.13.2 255.255.255.252
 no shutdown
 exit
interface Serial0/0/0
 description WAN Serial - to TCH-R2 S0/0/1 (this end is DTE)
 ip address 10.0.23.2 255.255.255.252
```

```
 no shutdown
 exit
!
ip dhcp excluded-address 192.168.30.1 192.168.30.20
ip dhcp pool LAB-LAN
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 exit
!
router rip
 version 2
 no auto-summary
 network 10.0.0.0
 network 192.168.30.0
 passive-interface GigabitEthernet0/0
 exit
!
ip domain-name school.local
username admin privilege 15 secret Admin@123
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
line console 0
 password cisco
 login
 logging synchronous
 exit
line vty 0 4
 login local
 transport input ssh
 exit
end
write memory
```

If `crypto key generate rsa` asks "How many bits in the modulus", type 1024. The passwords here (class123, cisco, Admin@123) are lab placeholders; change them if the routers are shared.

Switch configurations and end devices

The switches need no routing; they get a name, a management IP, a gateway and SSH. SW-LIB is shown in full. SW-TCH and SW-LAB are identical apart from the values in the table below (full versions are in the `.txt` files).

```
enable
configure terminal
hostname SW-LIB
no ip domain-lookup
service password-encryption
enable secret class123
banner motd # SW-LIB - Library switch. Authorized access only. #
!
interface GigabitEthernet0/1
 description Uplink to LIB-R1 G0/0
 exit
interface range FastEthernet0/1 - 4
 description Library end devices
 switchport mode access
 spanning-tree portfast
 exit
!
interface Vlan1
 ip address 192.168.10.2 255.255.255.0
 no shutdown
 exit
ip default-gateway 192.168.10.1
!
ip domain-name school.local
username admin privilege 15 secret Admin@123
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
line console 0
 password cisco
 login
 logging synchronous
 exit
line vty 0 15
 login local
 transport input ssh
 exit
end
write memory
```

Switch	hostname	VLAN 1 IP	ip default-gateway	Access ports
Library	SW-LIB	192.168.10.2	192.168.10.1	Fa0/1 - 4

Switch	hostname	VLAN 1 IP	ip default-gateway	Access ports
Teachers' room	SW-TCH	192.168.20.2	192.168.20.1	Fa0/1 - 4
Laboratory	SW-LAB	192.168.30.2	192.168.30.1	Fa0/1 - 5

End devices (Desktop tab > IP Configuration in Packet Tracer):

- PCs and laptops: select **DHCP**. Each should receive an address from .21 up, with the router as gateway.
- Printers: **Static**, x.x.x.10, mask 255.255.255.0, gateway x.x.x.1 for their room.
- AUTO-PC in the Lab: **Static**, 192.168.30.20, mask 255.255.255.0, gateway 192.168.30.1.

Build steps

The whole build takes about an hour in Packet Tracer; the same order works on the physical lab routers.

1. Place 3 routers (2901) and name them LIB-R1, TCH-R2, LAB-R3.
2. On each router: Physical tab > power off > drag **HWIC-2T** into the first empty slot > power on.
3. Place 3 switches (2960): SW-LIB, SW-TCH, SW-LAB.
4. Place the end devices from the addressing table: 3 PCs/laptops and 1 printer per room, plus AUTO-PC in the Lab.
5. Cable everything as in the cabling table. For serial links, click the DCE router first (LIB-R1 for the R1-R2 link, TCH-R2 for the R2-R3 link).
6. Open each router's CLI, answer **no** to the initial configuration dialog, then paste its config.
7. Paste each switch's config the same way.
8. Set IPs on the end devices (DHCP or static as listed).
9. Wait about 30 seconds for RIP to exchange updates, then run the verification checklist below.
10. Export the running configs: run `show running-config` on each device, copy the output into `LIB-R1.txt`, `TCH-R2.txt`, and so on. These exported files are what the repo needs.
11. Save the Packet Tracer file (`school-rip.pkt`) and share it with the team.

Verification checklist

The build is done when every router shows two RIP-learned LANs and any PC can ping any other room.

Command (on each router)	What you should see
`show ip interface brief`	Every used interface `up / up`
`show controllers serial 0/0/0`	On the DCE end: `DCE V.35, clock rate 64000`
`show ip protocols`	`Routing Protocol is "rip"`, send/receive version 2, `Automatic network summarization is not in effect`, G0/0 listed as passive
`show ip route`	Two `R` routes to the other rooms' LANs, plus an `R` route to the WAN link you are not on
`show ip rip database`	All 6 networks (3 LANs, 3 WAN links)
`debug ip rip`	`sending v2 update to 224.0.0.9` every ~30 s on both WAN interfaces; stop with `undebug all`

Expected `show ip route` (RIP part) when all links are up

```
LIB-R1
R    10.0.23.0/30 [120/1] via 10.0.12.2, Serial0/0/0
                  [120/1] via 10.0.13.2, GigabitEthernet0/1
R    192.168.20.0/24 [120/1] via 10.0.12.2, Serial0/0/0
R    192.168.30.0/24 [120/1] via 10.0.13.2, GigabitEthernet0/1

TCH-R2
R    10.0.13.0/30 [120/1] via 10.0.12.1, Serial0/0/0
                  [120/1] via 10.0.23.2, Serial0/0/1
R    192.168.10.0/24 [120/1] via 10.0.12.1, Serial0/0/0
R    192.168.30.0/24 [120/1] via 10.0.23.2, Serial0/0/1

LAB-R3
R    10.0.12.0/30 [120/1] via 10.0.13.1, GigabitEthernet0/1
                  [120/1] via 10.0.23.1, Serial0/0/0
R    192.168.10.0/24 [120/1] via 10.0.13.1, GigabitEthernet0/1
R    192.168.20.0/24 [120/1] via 10.0.23.1, Serial0/0/0
```

`[120/1]` means administrative distance 120 (RIP) and a metric of 1 hop. The far WAN link shows two equal 1-hop paths, so RIP installs both and load-balances.

Connectivity tests

- [ ] PC-LIB-1 pings PC-TCH-1 and PC-LAB-1
- [ ] PC-TCH-1 pings Printer-LAB (192.168.30.10)
- [ ] `tracert 192.168.20.10` from a Lab PC goes LAB-R3 > TCH-R2 (2 router hops)
- [ ] AUTO-PC can `ssh -l admin 192.168.10.1`, `192.168.20.1` and `192.168.30.1`
- [ ] Screenshots of `show ip route` from all 3 routers saved for the report

Handoff to the team

For Abdulrahman (Python / netmiko)

Router	SSH address	Username / password	netmiko device_type
LIB-R1	192.168.10.1	admin / Admin@123	cisco_ios
TCH-R2	192.168.20.1	admin / Admin@123	cisco_ios
LAB-R3	192.168.30.1	admin / Admin@123	cisco_ios

- The `admin` user is privilege 15, so scripts land straight in enable mode; the enable secret is `class123` if needed.
- netmiko cannot reach a Packet Tracer network from a real laptop. Run the scripts against the physical 2901s in the lab, or rebuild this design in GNS3 / EVE-NG with IOSv routers (same configs; check interface names).
- To capture `debug ip rip` over SSH, send `terminal monitor` first.

For Mouhamed (failure tests and analysis)

- Suggested test: on LIB-R1, `interface Serial0/0/0` > `shutdown`. LIB-R1 should then reach 192.168.20.0/24 via LAB-R3 with metric `[120/2]` instead of `[120/1]`.
- Bring the link back with `no shutdown` and record how long the original route takes to return.
- RIP default timers: update 30 s, invalid 180 s, holddown 180 s, flush 240 s (`show ip protocols` confirms them).

- Worth noting in the limitations section: RIP counts hops only, so it treats the Ethernet link (R1-R3) and the slower 64 kbps serial links as equal.